In simulation, given enough dimensions, the latent conditional model tends to learn near orthogonal representations regardless of graphical dependencies. One plausible rationale behind this phenomenon is the tendency for steering vectors in both unembedding and embedding spaces to possess large norms because, by Proposition 11, the inner product of embedding and unembedding steering vectors of the same concept is large. However, for steering vectors representing distinct concepts, it's often necessary for the inner product to be small as dictated by what's given in the training distributions. To accommodate for this, their cosine similarities tend to be close to zero. As a consequence, unembedding representations of different concepts often end up being near orthogonal as well.

Remark 9. It is worth noting that the conditions for orthogonal representations presented here can break in practice for LLMs. For instance, the injectivity assumption might not hold, and embedding and unembedding representations might not have perfectly matched representations.

5 Experiments

In this section, we present two slates of experiments to validate and augment our theoretical contributions. Specifically, in Section 5.1, we run experiments on simulated data from the latent conditional model to verify the existence of linear and orthogonal representations in this theoretical framework and how training on smaller dimensions, incomplete sets of contexts, and concept vectors can affect the representations. Then, we run experiments on large language models in Section 5.2 to show nontrivial alignment between embedding and unembedding representations of the same concept as predicted by our theory.

5.1 Simulated experiments

Simulation setup For simulation experiments, we first create simulated datasets by initially creating random DAGs (see Remark 1) with m variables/concepts. For a specific random DAG, the conditional probabilities of one variable given its parents are modeled by Bernoulli distributions where the parameters are sampled uniformly from $[0.3, 0.7]$. For example, suppose the DAG is $C_1 \to C_2$. Then $C_1 \sim \text{Bern}(p_1)$, $(C_2|C_1 = 0) \sim \text{Bern}(p_2)$, $(C_2|C_1 = 1) \sim \text{Bern}(p_3)$ where $p_1, p_2, p_3 \sim \text{Unif}([0.3, 0.7])$. In other words, we generate random graphical models wherein both the structures and distributions are created randomly. From these random graphical models, we can sample values of variables which are binary vectors as our datasets.

To let models learn conditional distributions, we train them to make predictions under cross-entropy loss after randomly masking the sampled binary vectors. Unless otherwise stated, the model is trained using stochastic gradient descent with a learning rate of 0.1 and batch size 100. More details are deferred to Appendix E.

Complete set of conditionals ($\widehat{\mathscr{C}} = \mathscr{C}, \widehat{\mathscr{D}} = \mathscr{D}$) We first run experiments where the model is trained to learn the complete set of conditional distributions. The representation dimension is set to be the same as the number of latent variables. Given a concept, to measure linearity, we calculate the average cosine similarities among steering vectors in the unembedding space and in the embedding space as well as between steering vectors in these two spaces. The result is shown in Table 1 which suggests that under the latent conditional model with the complete set of contexts, learned representations are indeed linearly encoded. In addition, Figure 5 in Appendix E shows that as loss decreases, the cosine similarities increase rapidly. Due to the exponential number of vectors that need to be trained, one cannot directly simulate a large number of latent variables. However, as we will show later (Appendix E), one can use incomplete sets $\widehat{\mathscr{C}}$ and $\widehat{\mathscr{D}}$ and simulate with more latent variables.

Table 1: When the model is trained to learn the complete set of conditionals, the m latent variables are represented linearly, and the embedding and unembedding representations are matched. The table shows average cosine similarities among and between steering vectors of unembeddings and embeddings. Standard errors are over 100 runs for 3 variables and 4 variables, 50, 20 and 10 runs for 5, 6 and 7 variables respectively.

m	Unembedding	Embedding	Unembedding and Embedding
3	0.972±0.006	0.982±0.005	0.980±0.005
4	0.975±0.005	0.971±0.005	0.973±0.005
5	0.988±0.004	0.981±0.004	0.984±0.004
6	0.997±0.000	0.985±0.002	0.990±0.001
7	0.995±0.001	0.972±0.004	0.981±0.003

Orthogonality Section 4 argues that, under the latent conditional model, unembedding representations of separated concepts will be orthogonal. In practice, Figure 7(a) (Appendix E) shows that this happens for simulated data even without latent variables being separated in the latent graph. We test the orthogonalities of representations in the latent conditional model by calculating the average cosine similarities between different sets of steering vectors. To see how well the theoretical framework fits LLMs, we similarly calculate the average cosine similarities between different sets of steering vectors in LLaMA-2 [Tou+23] using the counterfactual pairs in [PCV23] and they're reported in Figure 3. In particular, Figure 7 (Appendix E) shows that both simulated experiments and LLaMA-2 experiments exhibit similar behaviors where steering vectors of the same concept are more aligned while steering vectors of different concepts are almost orthogonal. This validates the claim that the latent conditional model is a suitable proxy to study LLMs.

In real-life datasets, not every concept or context vector has a natural correspondence to a token or sentence. On the other hand, the number of tokens and sentences grows exponentially with the number of latent concepts. Therefore, it is not realistic to always have $\widehat{\mathscr{D}} = \mathscr{D}$ or $\widehat{\mathscr{C}} = \mathscr{C}$. In other words, one does not need the next token prediction to learn all possible conditional probabilities perfectly. Because we are in a simulated setup, it's easy to probe the behavior in these situations. Therefore, we also perform additional experiments to observe these aspects of our simulated setup, in particular (i) Training on an incomplete set of contexts ($\widehat{\mathscr{D}} \subset \mathscr{D}$), (ii) Incomplete set of concept vectors ($\widehat{\mathscr{C}} \subset \mathscr{C}$). These are deferred to Appendix E. The experiments show that linearity is robust to these changes.

Moreover, in practice, the representation dimension is typically much smaller than the number of concepts represented. Thus, we run additional experiments in Appendix E with decreasing dimensions and observe one can still get reasonable linear representations.

Finally, we show in Appendix E that gradient descent or stochastic gradient descent is not the only algorithm that can induce linear representation, running other first-order methods like Adam [KB15] can lead to a similar pattern.

5.2 Experiments with large language models

In this section, we will conduct experiments on pre-trained large language models. We first emphasize that prior works [Elh+22; Bur+22; Tig+23; NLW23; Mos+22; PCV23; Gur+23] have already exhibited certain geometries of LLM representations related to linearity and orthogonality, via experiments. Therefore, in this section, we probe the geometry of the LLM representations, in particular that of the interplay between the embeddings and unmbeddings of the contexts and the tokens, that have not been explored in prior works. As [PCV23] remark, for a given binary concept, it's hard to generate pairs of contexts that

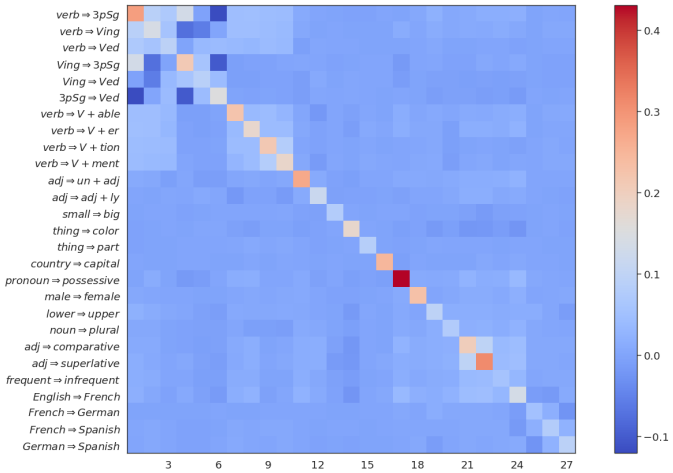

Figure 3: Unembedding steering vectors of the same concept in LLaMA-2 have nontrivial alignment, but steering vectors of different concepts are represented almost orthogonally.

differ in precisely this context, partly because of nuances of natural language and mainly because such counterfactual sentences are hard to construct even for human beings. In this section, we try to recreate these sets of missing experiments in the literature using existing open-source datasets.

Multilingual embedding geometry We first consider language translation concepts. For the embedding vectors, we consider pairs of contexts (x^0, x^1) where x^0, x^1 are the same sentences but in different languages. We consider four language pairs French–Spanish, French–German, English–French, and German–Spanish from the OPUS Books dataset [Tie12]. We take 150 random samples and filter out contexts with less than 20 or more than 150 tokens. For the unembedding concept vectors, we use the 27 concepts as described in [PCV23], which were built on top of the Big Analogy Test dataset [GDM16]. Examples of both datasets and a list of the 27 concepts are shown in Appendix F.

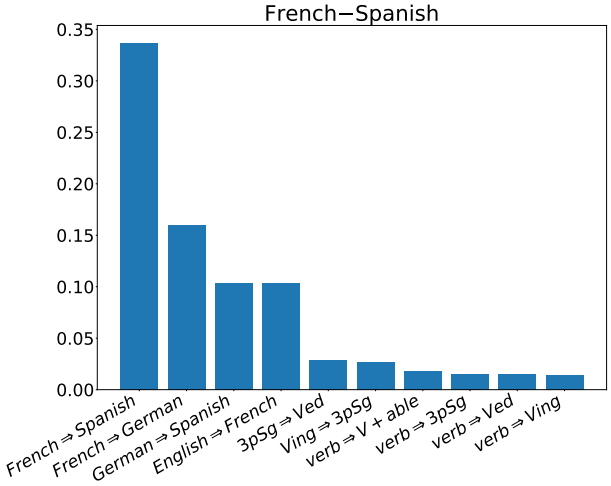

Figure 4: The French–Spanish concept is highly correlated with similar token concepts relative to others. Figure shows cosine similarities between the French–Spanish concept and token concepts.

We consider embeddings and unembeddings from LLaMA-2-7B [Tou+23]. We compute the absolute cosine similarity between the average embedding steering vector and the average